

ENGINEERING CASE STUDY: Empirical Volumetric Efficiency Validation

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Platform: 2018 Ford F-150 (2.7L Twin-Turbo V6 EcoBoost)	Software/Hardware: FORScan, MATLAB, OBDLink EX
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SITUATION

Modern forced-induction powertrains rely on precise absolute pressure delivery to maintain volumetric efficiency under varying ambient environmental loads. To establish a data-driven performance baseline for future component degradation studies, a rigorous evaluation of the engine's real-world dynamic response was required.

TASK

The objective was to capture high-resolution, multiplexed sensor data—specifically Manifold Absolute Pressure (MAP), Charge Air Temperature (IAT2), and Dual-Injection (PFDI) utilization—during a sustained, high-load operational envelope.

ACTION

Leveraging military-standard diagnostic discipline, I established a controlled field-testing protocol. I utilized a high-speed OBDLink multiplex adapter and FORScan telemetry software to log live Powertrain Control Module (PCM) data at 10 Hz during a 3rd-gear Wide Open Throttle (WOT) sweep from 2,500 to 5,000 RPM. The raw CSV datasets were subsequently processed using a custom MATLAB script to calculate the actual engine load profiles and plot the delta between the commanded and actual absolute pressure.

RESULT

Key Quantitative Results:

- Peak Absolute Manifold Pressure: 211 kPa (16.2 psi boost above 99 kPa atmospheric baseline)
- Peak Calculated Engine Load (VE Proxy): 98% during 3rd-Gear WOT Sweep
- PFDI System Mode: 89% Direct Injection / 11% Port Injection at 20,000 kPa Fuel Rail Pressure

F-150 2.7L EcoBoost: Wide Open Throttle (WOT) Data Analysis

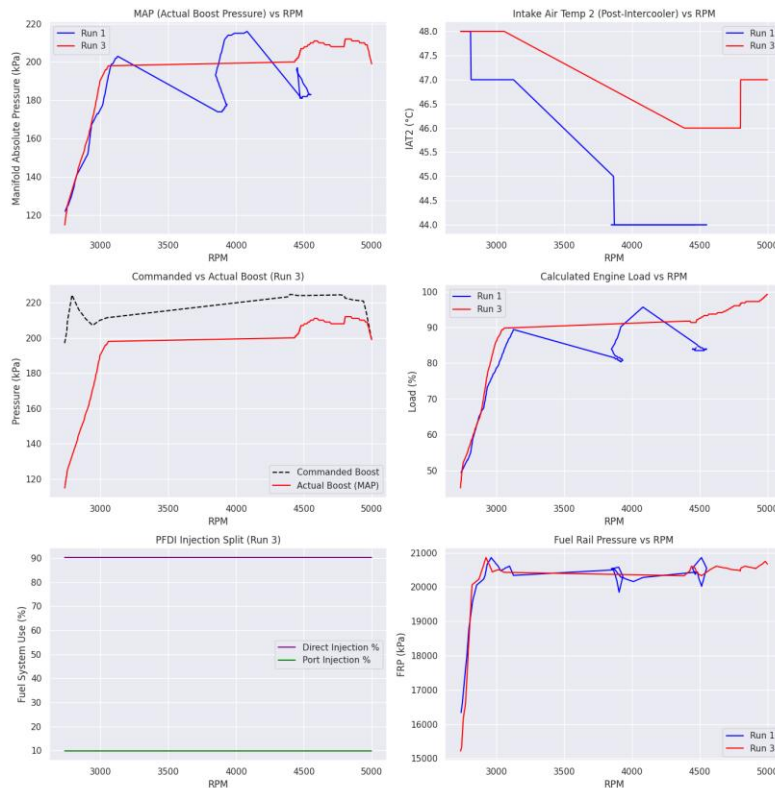


Figure 1.1: 6-Panel WOT Telemetry Analysis generated via MATLAB script *VE_Analysis_F150.m*.

DATA ANALYSIS & ENGINEERING FINDINGS

- Boost Delivery & Target Tracking:** Across the 3rd-gear WOT sweep, actual boost delivery (MAP) tracked closely with the PCM's commanded boost target through the midrange, but a measurable gap opened above 4,000 RPM, where commanded pressure climbed toward 225–230 kPa while actual MAP plateaued near 208–211 kPa. This ~15–20 kPa (roughly 2–3 psi) shortfall at the top of the pull indicates the twin-turbo system approaching its transient response ceiling under sustained high-load demand, consistent with the observed peak boost of 211 kPa (16.2 psi above the 99 kPa atmospheric baseline).
- Intercooler Thermal Rejection:** Post-intercooler charge air temperature (IAT2) rose progressively across the RPM sweep, climbing from a low of 44–46°C at moderate load into a heat-soak ceiling of approximately 47°C as boost demand and dwell time increased. This upward trend reflects the air-to-air intercooler's diminishing thermal margin under sustained WOT conditions, where heat rejection capacity is increasingly outpaced by compressor discharge temperature—an important baseline for evaluating intercooler efficiency degradation in future testing.
- Dual Injection Dynamics:** The PFDI (Port Fuel and Direct Injection) system maintained a consistent 89% direct injection / 11% port injection split throughout the WOT sweep, with fuel rail pressure holding steady near 20,000 kPa. This heavy bias toward direct injection at high load reflects the PCM's strategy of maximizing charge cooling and knock margin under boost, while

the minor port injection contribution supports injector duty-cycle balancing at the top of the RPM range.